

NODE-OPS-1.0

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Aziel Eliab • public work identity only

Node operations and security framework

NODE-OPS-1.0 • review of github.com/AzielEliab • 2026-09-06 • Aziel Eliab

Bulletproof here means smallest public surface, named refuses, isolate-then-phoenix, no auto-heal onto a

poisoned ID. It does not mean an unbreakable host. Phoenix is wait / re-seal after poison or isolation, not

public hostname resurrection. Sites pulled die with the pull.

1. Fleet review (2026-09-06)

Public repos already point the right way: FragGate hashed registry and refuse ledger; corpus security headers; runtime thin

MCP; qnm-node on 127.0.0.1 with APG, PHOENIX-LOCK wait, no account resurrection, no auto-heal; ARK

hosted-never-unlocks; azos/exec stub; MirageGrid hop stubs; AZMail smtp/login stubs. Gaps: BUILD-1.0 docs still say

bearers default-off (amend per QNM-WP-1.0: process ON, public rollup); cell-of-25

dual-bridge rotation not fleet-wired;

Q×act seal not yet the only legal success path on every fraggate_call.

2. Node classes

Class	Fleet	Public surface allowed
Face mirror. No Node Gate. No vault unlock.	azieleliab, corpus, godlock	Pages, cite, llms, software
Door GET /v1/mesh never enables.	fraggate, aziel-runtime	Thin MCP + fraggate_call.
Fabric public faces.	qnm-node	Loopback API only. Not on
Survival Prefer-central. Not a VPN.	azieltether	Counted download.
Overlay remote shell.	azos, azhub, azinterface	Status / place / cycles. No
Harness publish. No browser login.	azbot	Skill markdown. Queue
Vault unlock/scorch stub.	ark, EmbryoLock	Local only. Hosted
Assign stub.	miragegrid	assign live. mesh/vpn-hop
Side-net classify. Harvest/login/smtp stub.	aznet, azbrowser, azmail	Verify / cite-search /

3. Attack surface law

- ? One inbound door. FragGate or loopback. /p/{slug}/{op} is proxy, not exec.
- ? Allowlist ops only. Unknown slug = FG-HALLUC-TOOL. Stubs stay refuse.
- ? No credentials in process state. token_present only. No browser login.
- ? No public enable switch. GET does not change mesh state.
- ? qnm-node binds 127.0.0.1. Faces do not proxy that port.
- ? No vault on the Worker. No auto-heal to a spent mesh_id.
- ? No account resurrection. Memorial keeps the spend.
- ? Identity lock on export. Publish only via official API + AZBOT_PUBLISH=1.
- ? Local HTTP keeps nosniff, DENY frames, no-referrer, camera/mic/geo off, tight CSP.

4. Phoenix loop (failsafe)

- live ? APG miss or tamper ? isolate (cut tethers, spend mesh_id) ? memorial ?

PHOENIX-LOCK wait

(local, no controller hunt) ? declare comms clean ? assign new mesh_id ? rejoin as leaf of the

- local cell ? seat a new bridge pair if needed.

- ? This is not auto-heal. The old ID stays spent.

? Phoenix is wait / re-seal after poison or isolation. It is not ?bring the .uk node back.?

? Rejoin is local fabric only. Phoenix does not restore godlock.uk, azielcorpuslibrary.net, or any public rollup hostname.

? Sites pulled (token revoked, Worker dropped, DNS killed) ? public rollup on that hostname is down.

A local node may keep verifying and appending. Mesh does not climb back onto the public hostname by itself.

? A process supervisor restarting cloudflared is operator kit, not this paper. It fails if the credential or hostname is gone.

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? Phoenix does not un-scorch a vault.

? Phoenix does not resurrect a public hostname.

? Poison does not forward across cell edges.

? Three missed heartbeats = suspect. Suspect + APG hit = isolate.

? Heartbeat loss ? poison. Heartbeat loss ? ?apply last packet.?

? Phoenix is local reboot/WAIT for the failed node. Neighbors do not phoenix because a neighbor phoenix?d.

? Bridge poison rotates both bridges before inter-cell traffic.

? Held publish queue does not flush across phoenix without a new operator act.

4b. Split the wires

Fast 0.5?1s tick: presence + tip hash only. Fixed-size. No body, no diff, no ?also here?s the file.?

cite that

prev, match the lockset rule, and verify fail-closed. 777s is dwell after a valid cite, not ?wait then

take whatever arrived.? Clock desync is not a yes. Ambiguous tip is isolate, not merge.

Equivocation ends the peer, not the chain. Same prev, two different tips from one node ? that node

is locked/isolated. No vote-to-reconcile. Quorum cannot outvote a broken hash. Majority is not truth.

Emit last, locally. A node may announce a tip only after its own verify passes. No unsend, so nothing

leaving the box is an unverified body.

Partition: each island keeps its own chain; they do not auto-splice on reconnect. Rejoin is cite +

human/operator or lockset gate, same as first ingest.

The 1s loop and the 777s gate stay strangers. Anything less is a delayed epidemic.

Public tunnels/sites still die with the pull. Phoenix is not public hostname resurrection.

4c. Cold-copy survival

Multiply cold copies. Refuse live body sync across the network. A tip is content-addressed and

expensive to erase. Unkillable by single-server pull: that named hostname dies with the pull;

vaults that already hold the hashes keep verifying and appending. Hash-absolute poison refuse ?

equivocation isolates that peer, not the chain. Data outlives creators via content-addressed

tips + local verify/append. Payloads are pull-only and cold. Named hosts only.

4d. Re-expand-from-archive

Bytes survive, not summaries. Re-expand restores from archive after prev-hash verify.

It is not mesh from index. Crawlers are extra shelves only. Training residue is rumor.

Die-with-the-pull, split-the-wires, and cold-copy survival stay locked.

4e. REHEAL

A poisoned node heals from its own last good tip + a verified trusted pull, or phoenix-WAIT.

Never by listening to neighbors. Allowed: live / locked / isolated / tip-hash.

Forbidden: bodies / diffs / vote-to-fix.

Isolation is the cure. Neighbor talk-back-to-health is a group hug over a wound.

4f. CROSS-NETWORK-SURVIVAL-1.0

If network and data die tomorrow, the chain survives on cold shelves (hosts / DOI / git / vault).

Under that sentence sit die-with-the-pull, split-the-wires, cold-copy survival, re-expand-from-archive, and REHEAL. The live mesh is not a shelf.

5. Per-class refuse (minimum)

Class	Must refuse
Face	Node Gate, IP panel, vault unlock, mesh enable
Door	Invented slugs, stub ops, runtime_run as agent default
Fabric bind	cell_full join, leaf as inter-cell path, spent ID reuse, public
Survival is down	vpn / arm / mesh-join stubs; opening a public mesh because central
Overlay	azos/exec; exec of catalog engines from Interface

Vault	hosted unlock / encrypt / scorch / wipe
Assign	mesh, vpn-hop, hop, tunnel
Side-net	harvest, login, smtp send, invented visits, deanonymize
Engine	any op not on the 1.6.13 allowlist

6. Receipts

Every node writes AZL-LEDGER kinds: boot, heartbeat, refuse, isolate, phoenix, assign, rotate, tether_cut, hold. No model sentence. No token body.

7. Close tests

- ? Public faces have no enable switch and no vault unlock.
- ? Door stubs refuse and the refuse is a receipt.
- ? qnm-node is not reachable off loopback in default config.
- ? Spent mesh_id rejected on rejoin. Phoenix wait has no outbound hunt packet.
- ? Phoenix wait / re-seal does not restore godlock.uk or reattach a public hostname.
- ? Tick plane refuses body / diff / file. Payload is pull-only. Equivocation isolates that peer.
- ? 777s dwell is after a valid cite. Clock desync is not a yes. No auto-splice on reconnect.
- ? Cold copies multiply. Live body sync refused. Single-server pull does not erase local tips.
- ? Held queue does not flush across phoenix without a new act.
- ? ARK hosted unlock remains stub. Cell does not halt when one leaf isolates.

Does not add a Node Gate to ?see security.? Does not turn MirageGrid into a VPN. Does not publish exploit recipes. Public identity: Aziel Eliab only.

If the files hold, the name was never the point.

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